

PANTHER JUNCTION 2 N, TX, USA

WMO: 747280

Lat: 29.3483N

Lon: 103.2093W

Elev: 1065

StdP: 89.17

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 22016

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.3	(c) -1.4	(d) -18.8	(e) 0.8	(f) 16.7	(g) -16.4	(h) 1.0	(i) 17.1	(j) 5.7	(k) 19.4	(l) 5.0	(m) 17.4	(n) 1.4	(o) N/A	(p) 0.361

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	14.3	38.0	17.7	36.8	17.7	35.7	17.7	21.3	29.0	20.8	28.6	20.4	28.6	2.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	16.6	22.8	19.0	15.7	22.9	18.3	15.0	23.2	67.2	28.7	65.4	28.4	63.8	28.4	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.9	4.4	3.9	-6.8	40.4	2.7	1.0	-8.7	41.1	-10.3	41.7	-11.9	42.2	-13.9	42.9		
			-8.2	22.3	2.8	0.5	-10.3	22.7	-11.9	23.0	-13.5	23.2	-15.5	23.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.9	9.7	13.1	16.8	21.0	24.8	28.1	27.8	27.8	24.3	20.4	14.1	10.6
	DBStd	7.67	4.82	5.40	4.76	3.90	3.82	2.51	2.27	2.41	2.79	4.04	4.57	4.49
	HDD10.0	167	66	28	8	1	0	0	0	0	0	1	16	48
	HDD18.3	937	270	156	84	19	4	0	0	0	2	27	136	239
	CDD10.0	3785	55	114	218	331	459	544	551	552	429	322	141	68
	CDD18.3	1513	1	9	36	99	205	294	293	294	181	91	10	1
	CDH23.3	19293	36	198	625	1424	2664	3937	3575	3732	1828	1041	194	39
	CDH26.7	9608	2	47	197	627	1411	2248	1878	1993	776	392	35	1
(14) Wind	WSAvg	1.8	1.6	1.8	1.9	2.1	2.2	2.1	2.1	1.9	1.6	1.6	1.5	1.6
(15) Precipitation	PrecAvg	335	12	12	10	13	32	43	65	58	37	31	15	9
	PrecMax	523	69	58	53	43	88	105	126	159	93	117	56	67
	PrecMin	177	0	0	0	0	0	8	15	12	0	0	0	0
	PrecStd	87	15	14	12	12	19	25	35	35	31	33	15	14
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.2	30.9	32.5	35.6	38.8	40.2	38.2	38.2	35.6	33.4	29.7	26.6
		MCWB	11.1	12.2	12.7	13.8	16.3	17.2	18.6	19.0	18.0	16.0	13.4	11.5
	2%	DB	24.1	28.0	30.7	33.8	36.7	38.6	36.8	36.7	34.0	32.2	27.5	24.4
		MCWB	10.0	11.5	12.4	13.4	15.2	17.5	18.5	18.7	17.7	16.3	12.9	10.6
	5%	DB	22.0	25.7	29.0	32.3	35.2	37.2	35.6	35.7	32.6	30.7	25.6	22.5
		MCWB	9.4	10.8	12.0	12.9	14.8	17.7	18.4	18.7	17.9	16.1	12.7	10.5
	10%	DB	19.5	23.6	27.2	30.5	33.6	35.7	34.3	34.5	31.1	29.0	23.7	20.3
		MCWB	8.5	10.4	11.7	12.7	15.2	18.1	18.5	18.7	17.9	15.8	12.4	9.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.3	14.8	16.7	18.2	20.9	21.7	22.4	21.8	21.5	20.3	17.2	13.5
		MCDB	18.4	24.2	25.3	26.9	30.3	30.2	28.9	30.8	28.0	28.5	22.6	21.1
	2%	WB	11.7	13.6	15.3	17.0	20.0	20.8	21.3	21.1	20.8	19.4	15.8	12.4
		MCDB	20.9	22.9	23.7	25.4	29.0	29.9	27.8	29.5	27.9	26.8	22.2	20.7
	5%	WB	10.3	12.3	14.2	16.2	19.1	20.4	20.7	20.6	20.4	18.6	14.7	11.4
		MCDB	20.7	22.0	23.3	24.9	27.8	29.8	27.7	29.2	27.3	25.8	21.9	19.6
	10%	WB	9.0	11.1	13.2	15.2	18.5	19.9	20.2	20.1	20.0	17.7	13.3	10.2
		MCDB	18.8	22.1	22.7	24.2	27.1	29.7	28.3	29.2	26.8	24.9	21.3	18.2
(35) Mean Daily Temperature Range	5% DB	MDBR	14.6	15.9	16.4	16.2	15.1	14.3	12.3	12.4	11.9	14.4	14.7	14.4
		MCDBR	17.6	18.4	18.8	17.8	16.5	15.9	13.8	13.6	14.2	16.0	17.2	17.3
		MCWBR	8.3	7.9	7.5	6.9	6.7	4.4	3.5	3.4	4.0	5.8	7.5	7.7
	5% WB	MCDBR	16.6	17.2	16.5	15.7	15.1	13.9	11.5	12.2	10.8	12.9	15.0	15.4
		MCWBR	8.2	7.9	6.5	6.2	6.3	4.0	3.4	3.4	3.2	4.9	7.2	7.4
(40) Clear-Sky Solar Irradiance	taub		0.262	0.267	0.281	0.319	0.352	0.367	0.390	0.366	0.358	0.302	0.277	0.263
	taud		2.659	2.627	2.570	2.441	2.383	2.417	2.393	2.482	2.437	2.611	2.616	2.634
	Ebn at Noon		998	1015	1012	974	935	915	894	916	916	960	968	979
	Edh at Noon		76	86	97	115	122	118	120	109	110	86	79	75
(44) All-Sky Solar Radiation	RadAvg		3.80	4.79	6.00	6.97	7.42	7.43	6.87	6.50	5.66	4.95	4.07	3.54
	RadStd		0.36	0.42	0.30	0.38	0.31	0.33	0.40	0.41	0.36	0.41	0.35	0.24

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+223	+154

Nomenclature: See separate page